

LAUREN LIU

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EDUCATION

University of California, Santa Cruz
B.S. Computer Science with Honors

Graduation: June 2026
Cum Laude

RELEVANT PROJECTS

PepTrack App (SwiftUI, Firebase, USDA API, Groq API) | [Demo](#)

Fall 2025 - Winter 2026

- Built a full-stack iOS nutrition tracking app in a team of 5 using Agile/Scrum methodology
- Led UI development across dashboard, onboarding, goals, profile, and progress views in SwiftUI
- Integrated Firebase Authentication and Firestore for user accounts, food logging, and real-time data sync
- Implemented food search via the USDA FoodData Central API; built macro tracking, XP/leveling, streak tracking, hydration goals, and AI-powered meal suggestions via Groq API

Computer Graphics Projects (WebGL) | [Demo](#)

Winter 2026

- Built progressively complex WebGL scenes including an interactive drawing tool, a first-person Minecraft-style world, and a 3D space scene with 20+ unique geometry types and imported GLTF models
- Implemented dynamic lighting with 3+ light types, mouse-look camera, block placement/removal, procedural terrain generation, and an animated day/night cycle

HTTP Server (C, POSIX)

Fall 2025

- Built a multithreaded HTTP server from scratch using POSIX sockets and a thread pool, supporting concurrent connections, GET/POST requests, and static file serving with robust error handling

Computer Systems Projects (C, Unix, Valgrind)

Fall 2023

- Implemented sorting algorithms, abstract data types (Set, linked list), and a numerical library in C; managed memory manually with malloc/free, debugged with Valgrind, and built with Unix tools and Makefiles

WORK EXPERIENCE

University of California, Santa Cruz, Reader

Spring 2025 - June 2026

- Served as Reader for four CS courses: CSE 20 (Python), CSE 102 (Data Structures and Algorithms), CSE 120 (Computer Architecture), and CSE 109 (Quantum Computing)
- Ensured objective, fair evaluation across assignments, projects, and exams - maintaining grading consistency across classes of 60–300 students
- Developed grading rubrics in Gradescope and coordinated with instructors on edge cases across topics including RISC-V, cache memory, algorithm proofs, and quantum computing

Brainly | Super Moderator, Tutor

Summer 2018 - Fall 2025

- Managed a team of 30 tutors in math, physics, and CS; trained new members, evaluated peers, and ensured site safety for a student audience

City of Anaheim | Software Engineering Intern

Summer 2022

- Collaborated with city officials to identify and analyze infrastructure inefficiencies in pooling and irrigation systems, proposing data-driven solutions to improve drainage throughput

EXTRACURRICULARS

Sigma Omicron Pi, Secretary, Academic Chair, Sergeant-at-Arms, VP Philanthropy

Winter 2023 - June 2026

- Led weekly business meetings, managed cabinet agendas, and ensured compliance with national requirements
- Planned and executed a multi-chapter philanthropy event including dog toy drives, food/clothing donations to a local shelter, and volunteer work with Second Harvest Food Bank
- Promoted academic engagement through study groups, monthly presentations, and peer mentorship

Women In Science And Engineering (WISE)

Fall 2022 - Spring 2023

- Organized leadership workshops and mentored new members; competed in the Santa Cruz Hackathon (CruzHacks)

SKILLS

Python, Java, C, C++, Assembly, HTML, CSS, Javascript, LaTeX, Data Analysis, Machine Learning, Microsoft Office (Excel, Powerpoint, Word), Swift, SwiftUI, Firebase/Firestore, SQL, Xcode, Git, REST APIs, iOS Development, WebGL, Three.js, GLSL, Computer Graphics